BAHRAIN POLYTECHNIC بوليتكنك البحرين

Assessment Cover Sheet

Assessment	Project (Group)		
Due Date	04/01/2026	**Course Code**	IT8414
Course Title	Deep Learning for Data Analytics		
Assessment	Uncontrolled	Individual	Not must-pass
Internal Moderator's	Mr. Gus Finucane		
External Examiner's Name	Dr.Mekala Soosay		

Instructions:

1. This cover sheet must be completed (section in blue below) and attached to your assessment before submission in hard copy/soft copy.
2. The time allowed for this assessment is 6 - 8 weeks.
3. This assessment carries 100 marks distributed to a total of 1 report with Viva/Demo assessing CILO 1, 2 and CILO 3.
4. The **use of generative AI tools is restricted to knowledge building, coding support and data generation**.
5. References consulted (if any) must be properly acknowledged and cited.

Learner ID		**Date Submitted**	
Learner Name			
Programme	ICT8040/ICT8041		
Programme	Bachelor of ICT (Databases)		
Lecturer's Name	Dr. Shomona Jacob		

By submitting this assessment for marking, I affirm that this assessment is my own work.

Learner	

BAHRAIN POLYTECHNIC بوليتكنك البحرين

Assessment 2 — Group Project

IT8414 Deep Learning for Data Analytics · Brief, Marking Rubric and Templates

Academic Year 2026/2027 · Semester 1

Table of Contents

IT8414 – Deep Learning for Data Analytics 3
Assessment 2: Group Project — Brief, Marking Rubric and Report Template 3
1. Course Intended Learning Outcomes assessed 3
2. Project aim 4
3. Choosing your application 4
4. Required work 5
Part A — Problem definition and background (CILO 1) 5
Part B — Data preparation (CILO 2) 5
Part C — Model design and implementation (CILO 2) 5
Part D — Training and optimisation (CILO 2) 5
Part E — Evaluation and analysis (CILO 3) 5
Part F — Interpretation, limitations and ethics (CILO 3) 6
Part G — Teamwork, live demonstration and individual viva (30 marks — see Section 5) 6
5. Teamwork, demonstration and viva (30 marks) 6
5.1 Why this component is weighted so heavily 6
5.2 Format 6
5.3 The live modification task 7
5.4 Individual viva question bank (indicative) 7
5.5 Individual marking 7
6. Deliverables 8
7. Milestones 8
8. Marking rubric (100 marks) 9
Band descriptors 9
Individual marks 14
9. Report template 14
Notebook structure template 15
10. Contribution statement template 16
11. Project proposal template (Week [6], formative) 16
12. Policies 17
Pre-submission checklist 17

IT8414 – Deep Learning for Data Analytics

Assessment 2: Group Project — Brief, Marking Rubric and Report Template

Item	Detail
Course code / title	IT8414 – Deep Learning for Data Analytics (Level 8, 15 credits)
Assessment type	Project (Group) — Uncontrolled Assessment
Weighting	35% of the overall course grade
Marks out of	100
Pass mark	60%
Due	Week 13 (exact date and time to be confirmed by the tutor)
CILOs assessed	CILO 1 (20%), CILO 2 (30%), CILO 3 (50%)
Tools	Python, Jupyter Notebook, TensorFlow and/or PyTorch
Generative AI	Strictly prohibited unless explicitly permitted in writing by the tutor
Live defence	**30 of the 100 marks are awarded at the live demonstration and individual viva. These marks cannot be earned from the submitted artefacts alone.**

Note to students: this brief is deliberately generic. You choose the application domain, the dataset and the architectures. Every section below tells you *what* must be present; *how* you satisfy it depends on the problem you select. Where you see `[ . . . ]`, replace it with your own content.

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1. Course Intended Learning Outcomes assessed

#	Learning outcome	Weight in this assessment
CILO 1	Demonstrate a critical understanding of deep learning architectures and training algorithms.	20 marks
CILO 2	Develop deep learning solutions to address complex problems using modern frameworks.	30 marks
CILO 3	Use a range of approaches to analyse and evaluate deep learning models for solving real-world problems.	50 marks

Employability skills developed: communication, teamwork, problem solving, planning and organising, learning, technology.

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2. Project aim

Working in a group, you will select a **real-world classification problem**, obtain or construct a suitable dataset, and **design, implement, train, tune and critically evaluate deep learning models** that solve it. You will justify your architectural choices against the theory covered in the course, compare at least two model configurations, and communicate your findings in a technical report, a live demonstration and an individual viva.

The emphasis is not on achieving the highest possible accuracy. It is on **sound methodology, defensible design decisions, honest evidence-based evaluation, and your ability to explain and modify your own work without assistance**.

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3. Choosing your application

You may propose any classification problem, provided it is non-trivial and the data is legally and ethically obtainable. Indicative domains (choose one or propose your own):

Data type	Example problems	Typically suitable architectures
Images	Medical image screening, defect detection, plant disease identification, face or gesture recognition, waste sorting	CNN, transfer learning (VGG/ResNet/EfficientNet), Vision Transformer
Text	Sentiment analysis, spam or fake-news detection, ticket routing, Arabic/English topic classification	Embeddings + RNN/LSTM/GRU, 1-D CNN, Transformer / fine-tuned BERT
Audio / speech	Spoken command recognition, environmental sound classification, speaker identification	Spectrogram + CNN, CRNN, LSTM
Tabular	Credit default, churn, fraud, disease risk, student outcome prediction	Deep MLP, embeddings for categorical features, TabNet
Time series / sensor	Human activity recognition, machine fault detection, anomaly detection	1-D CNN, LSTM, hybrid CNN-LSTM

Constraints on the problem

- It must be a **classification** task (binary, multi-class or multi-label).
- The dataset must have enough samples to train and evaluate meaningfully, and must be split into training, validation and test sets.
- Two groups in the same class may not submit the same problem *and* the same dataset.
- Your problem must be approved by the tutor at the proposal stage (see Section 10).

4. Required work

Part A — Problem definition and background (CILO 1)

- State the problem, why it matters, and who the end user is.
- Describe the dataset: source, licence, size, number of classes, class balance, features, known limitations.
- Review at least **three credible sources** (papers, technical reports, documentation) describing how similar problems have been solved.
- Explain the theory of the architecture(s) you selected: how the layers work, what the activation and loss functions do, and which optimiser and training algorithm you use and why.

Part B — Data preparation (CILO 2)

- Cleaning, handling of missing or corrupt records, encoding, normalisation or standardisation.
- Type-appropriate preparation: image resizing and augmentation, tokenisation and padding, spectrogram extraction, windowing, and so on.
- A justified train / validation / test split, with any stratification or class-imbalance treatment explained.
- Exploratory analysis with at least two visualisations that inform a later design decision.

Part C — Model design and implementation (CILO 2)

- Implement a **baseline model** and at least one **improved or alternative model**. Acceptable pairings include: shallow vs deep, custom CNN vs transfer learning, RNN vs Transformer, or the same architecture with substantially different regularisation.
- Present the architecture of each model (layer table or diagram) with parameter counts.
- Code must run end-to-end in a Jupyter Notebook, with a fixed random seed and clear, commented cells.

Part D — Training and optimisation (CILO 2)

- Report training configuration: batch size, epochs, learning rate, optimiser, loss function, callbacks.
- Tune at least **three hyperparameters** and record the experiments in a table.
- Demonstrate why or why not - you detected and addressed overfitting or underfitting (learning curves plus the technique applied — dropout, early stopping, regularisation, augmentation, or similar).

Part E — Evaluation and analysis (CILO 3)

- Evaluate on the **validation/test set** using metrics appropriate to your problem: accuracy plus precision, recall, F1 (per class and averaged), confusion matrix, and ROC-AUC or PR-AUC where relevant.

- Justify why those metrics suit your problem — particularly if classes are imbalanced.
- Compare all models in a single results table and state which you would deploy, and why.
- Perform at least one form of **error analysis**: inspect misclassified samples, per-class weaknesses, or confusion patterns, and explain the likely cause.

Part F — Interpretation, limitations and ethics (CILO 3)

- Use visualisation or interpretability techniques appropriate to your data (Grad-CAM, saliency maps, attention weights, feature importance, embedding projections, or clear diagnostic plots).
- Discuss limitations, bias in the data, generalisation risk, and any ethical, privacy or deployment considerations.
- Recommend concrete future work.

Part G — Teamwork, live demonstration and individual viva (30 marks — see Section 5)

This is the largest single component of the assessment. It is assessed in the room, individually, and it spans all three CILOs.

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5. Teamwork, demonstration and viva (30 marks)

5.1 Why this component is weighted so heavily

Deep learning code is now easy to obtain and easy to generate. Understanding it is not. This component exists to verify that the work you submit is work you can **explain, defend and change on the spot**. A group that submits an excellent notebook it cannot explain will fail this assessment.

5.2 Format

Each group is allocated **[25–30] minutes**, structured as follows:

Stage	Duration	What happens	Marked as
1. Group presentation	[8] min	All members present. Each member presents a technical portion of the work.	C12
2. Live demonstration	[5] min	Run the notebook (or a saved-model inference path) on the tutor's chosen input. Show a prediction end to end.	C7
3. Live modification task	[5] min	The tutor names **one small change** to make in the notebook, and the team implements it live. See 5.3.	C7

Stage	Duration	What happens	Marked as
4. Individual viva	[2–3] min per member	Each member answers questions individually. Questions may cover **any** part of the project, not only the part they wrote.	C3, C7
5. Process and teamwork review	[3] min	Work log, commit history and contribution statement reviewed against what was observed.	C12

5.3 The live modification task

The tutor will ask for a change of the kind you would make routinely if the work is yours. Examples of the type (the actual task is not released in advance):

- change the activation function in a named layer and re-run that cell;
- add or remove a dropout layer and state what you expect to happen before running;
- change the evaluation to report per-class recall instead of overall accuracy;
- alter the batch size or learning rate and explain the effect on the curves;
- swap the loss function and explain why it is or is not appropriate here;
- plot the confusion matrix for a different data split.

You may use your own notebook and the official framework documentation. You may **not** use any AI assistant, prepared snippets from outside the notebook, or help from teammates during your turn.

5.4 Individual viva question bank (indicative)

Prepare for questions such as:

- Walk me through what happens to one sample as it passes through your network.
- Why this optimiser, and what would change if you used a different one?
- Point to the line where overfitting is being controlled, and explain the mechanism.
- Your validation accuracy is higher than your training accuracy in epoch 3 — why?
- Why is accuracy a poor metric for your dataset? Which metric did you rely on and why?
- What does this figure actually show, and what did you conclude from it?
- Which part of this code did you write, and what did you change from the source you adapted?

5.5 Individual marking

Marks for C3, C7 and C12 are recorded **per member**, not per group. A member who cannot explain the submitted work, or who cannot complete a routine modification, is marked accordingly regardless of the quality of the artefacts. Absence from the demonstration without an approved extenuating-circumstances claim results in **zero for all 30 marks** of this component.

6. Deliverables

Submit a single ZIP file named `IT8414_Group[NN]_[ShortProjectTitle].zip` containing and place the github link with the below contents.

#	Deliverable	Format / limit
1	Technical report	PDF, 2,000–3,000 words excluding references, tables, figures and appendices
2	Jupyter Notebook	`.ipynb` **with all outputs saved**, plus an HTML or PDF export
3	Dataset	The data file, or a `DATA.md` with the download link, licence and preparation script if the file is large
4	Presentation slides	PDF or PPTX, 10–12 slides
5	Contribution statement	Signed by all members (template in Section 9)
6	Work log / commit history	Weekly log or exported version-control history — required evidence for C12
7	`README.md`	Environment, library versions, and how to reproduce the results

Bring a working machine to the demonstration with the environment already installed and the dataset available offline. Environment failures on the day are not grounds for a re-sit.

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7. Milestones

Week	Milestone	Submission
Week [6]	Group registration and one-page project proposal: problem, dataset, candidate architectures, member roles	Moodle — formative, tutor approval required
Week [9]	Progress check: data prepared, baseline model trained, initial results. **Each member answers one question on the code shown.**	In class — formative, rehearsal for the viva
Week [12]	Draft report section headings and results table	Optional tutor feedback
Week 13	**Final submission, demonstration and individual viva**	**Moodle + in class — summative, 100 marks**

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8. Marking rubric (100 marks)

Marks are awarded per criterion using the band descriptors below. The pass mark for this assessment is **60%**.

#	Criterion	CILO	Marks	Assessed from
C1	Problem definition, dataset understanding and background research	1	6	Report
C2	Critical understanding of architectures and training algorithms	1	6	Report
C3	**Individual viva — theoretical defence**	**1**	**8**	**Live**
C4	Data preprocessing and feature engineering	2	6	Notebook + report
C5	Model design and implementation in a deep learning framework	2	8	Notebook
C6	Training, hyperparameter tuning and overfitting control	2	6	Notebook + report
C7	**Live demonstration, code modification task and implementation viva**	**2**	**10**	**Live**
C8	Evaluation methodology and use of metrics	3	12	Report + notebook
C9	Experimentation and comparative analysis	3	10	Report + notebook
C10	Interpretation, visualisation, limitations and ethics	3	9	Report
C11	Report structure, technical writing and referencing	3	7	Report
C12	**Teamwork, project management and presentation delivery**	**3**	**12**	**Live + work log**
	Total		**100**	

Teamwork and demonstration total: C3 + C7 + C12 = 30 marks.

CILO totals are preserved as mandated by the course descriptor: CILO 1 = 6+6+8 = 20; CILO 2 = 6+8+6+10 = 30; CILO 3 = 12+10+9+7+12 = 50.

Band descriptors

C1 — Problem definition, dataset understanding and background research (6) — CILO 1

Band	Descriptor
Excellent (85–100%)	Problem precisely scoped with a clear end user and value proposition. Dataset characterised in depth (provenance, licence, class balance, known biases, limitations). Four or more credible sources synthesised and directly informing the design.
Very good (70–84%)	Problem and significance clear. Dataset well described including class distribution and limitations. Three credible sources reviewed and connected to the chosen approach.
Pass (60–69%)	Problem stated adequately. Basic dataset description. Some relevant sources cited but the link to the design is loose or descriptive.
Fail (0–59%)	Problem vague or trivial. Dataset barely described. Sources absent, irrelevant or unreliable.

C2 — Critical understanding of architectures and training algorithms (6) — CILO 1

Band	Descriptor
Excellent (85–100%)	Accurate, confident written explanation of how the architecture processes the data, with correct treatment of activation functions, loss, gradient-based optimisation and regularisation. Alternatives compared; choice defended on principled grounds.
Very good (70–84%)	Correct explanation with minor gaps. Reasonable justification against at least one alternative.
Pass (60–69%)	Mostly correct but descriptive; reads as summarised documentation. Justification thin.
Fail (0–59%)	Significant conceptual errors, or architecture used with no explanation of why it fits the problem.

C3 — Individual viva: theoretical defence (8) — CILO 1 *(marked per member)*

Band	Descriptor
Excellent (85–100%)	Answers theory questions fluently and correctly across the whole project, not only their own section. Traces a sample through the network, explains the role of each design choice, and reasons confidently about counterfactuals ("what if we changed X"). Corrects or qualifies the tutor's premise where appropriate.
Very good (70–84%)	Answers most theory questions correctly with minor prompting. Clear grasp of the architecture and training algorithm used; some hesitation on parts written by teammates.
Pass (60–69%)	Answers basic questions about their own section correctly but relies on recall rather than understanding. Struggles with counterfactuals or with parts written by others.
Fail (0–59%)	Cannot explain core mechanisms of the submitted work, gives answers contradicting the report, or defers repeatedly to teammates.

C4 — Data preprocessing and feature engineering (6) — CILO 2

Band	Descriptor
Excellent (85–100%)	Complete, reproducible pipeline; every step justified against data type and model requirements. Imbalance, leakage risk and split design handled explicitly. EDA clearly drives later decisions.
Very good (70–84%)	Sound pipeline with appropriate scaling/encoding/augmentation and a correct stratified split. Most steps justified.
Pass (60–69%)	Adequate preprocessing that allows training, but steps unjustified or minor issues remain (e.g. imbalance unaddressed).
Fail (0–59%)	Preprocessing missing, incorrect, or data leakage between splits.

C5 — Model design and implementation (8) — CILO 2

Band	Descriptor
Excellent (85–100%)	Two or more well-constructed models, correctly implemented, architectures documented (layer tables/diagrams, parameter counts). Notebook clean, modular, commented, seeded, runs end to end.
Very good (70–84%)	Baseline and improved model both implemented correctly and documented. Code runs and is readable, with some structural weaknesses.
Pass (60–69%)	One functioning model plus a limited variant. Code runs but is disorganised, sparsely commented or adapted without modification.
Fail (0–59%)	Model does not train, only a single trivial model exists, or code cannot be reproduced.

C6 — Training, hyperparameter tuning and overfitting control (6) — CILO 2

Band	Descriptor
Excellent (85–100%)	Systematic tuning of three or more hyperparameters recorded in a comparison table with observations. Learning curves interpreted correctly; regularisation demonstrably improves generalisation.
Very good (70–84%)	Several hyperparameters varied and recorded. Overfitting identified from curves and a reasonable technique applied.
Pass (60–69%)	Limited tuning, informally reported. Overfitting mentioned but treatment superficial.
Fail (0–59%)	No tuning; training configuration unreported; overfitting ignored or misread.

C7 — Live demonstration, code modification task and implementation viva (10) — CILO 2
(marked per member)

Band	Descriptor
Excellent (85–100%)	Demonstration runs cleanly on tutor-chosen input. The modification task is completed quickly and correctly, with the expected effect stated *before* running and the actual

Band	Descriptor
	result interpreted afterwards. Navigates the codebase fluently, explains any line on request, and distinguishes clearly between adapted and original code.
Very good (70–84%)	Demonstration works with minor issues. Modification completed correctly, possibly with a hint or a documentation lookup. Can explain most of the code confidently.
Pass (60–69%)	Demonstration works. Modification attempted and substantially correct but slow, requiring several prompts. Explains their own cells but is uncertain elsewhere in the notebook.
Fail (0–59%)	Demonstration fails through lack of preparation, or the modification cannot be completed, or the member cannot locate and explain the code they claim to have written.

C8 — Evaluation methodology and use of metrics (12) — CILO 3

Band	Descriptor
Excellent (85–100%)	Rigorous protocol on a genuinely held-out test set. Well-chosen metrics, each justified in terms of the problem's costs and class balance. Confusion matrix and per-class results interpreted, not merely displayed.
Very good (70–84%)	Correct test-set protocol; multiple appropriate metrics reported and mostly interpreted. Justification present but brief.
Pass (60–69%)	Evaluation performed but relies mainly on accuracy, or interpretation restates numbers without insight.
Fail (0–59%)	Evaluated on training or validation data, metrics inappropriate, or results not reported.

C9 — Experimentation and comparative analysis (10) — CILO 3

Band	Descriptor
Excellent (85–100%)	Well-designed comparison across models and configurations in a clear results table. Differences explained causally; error analysis identifies specific failure modes; deployment recommendation follows from the evidence.
Very good (70–84%)	Models compared fairly with a results table and sensible explanation of differences. Some error analysis performed.
Pass (60–69%)	Comparison present but shallow — results side by side with little explanation of why they differ.
Fail (0–59%)	No meaningful comparison, or conclusions contradict the evidence.

C10 — Interpretation, visualisation, limitations and ethics (9) — CILO 3

Band	Descriptor
Excellent (85–100%)	Interpretability technique well matched to the data type and genuinely informative. Figures clear, labelled and purposeful. Limitations, dataset bias, generalisation risk and ethical/deployment concerns discussed with insight, leading to specific future work.
Very good (70–84%)	Useful interpretation and good-quality visuals. Limitations and ethical issues addressed with reasonable depth.
Pass (60–69%)	Basic plots included; limitations acknowledged generically. Ethics treated as a formality.
Fail (0–59%)	Figures missing, unlabelled or irrelevant. No discussion of limitations or ethics.

C11 — Report structure, technical writing and referencing (7) — CILO 3

Band	Descriptor
Excellent (85–100%)	Professional report: logical structure, precise technical language, all figures and tables numbered, captioned and referenced in the text. Consistent citation style. Within word limit.
Very good (70–84%)	Well organised and readable with consistent referencing and minor lapses in captioning or style.
Pass (60–69%)	Structure present but uneven; some figures uncaptioned, referencing inconsistent, or expression unclear in places.
Fail (0–59%)	Disorganised, substantially over or under length, or referencing absent.

C12 — Teamwork, project management and presentation delivery (12) — CILO 3 *(marked per member)*

Band	Descriptor
Excellent (85–100%)	Work log and commit history show planned, evenly distributed, sustained work across the semester by all members. Presentation is well structured, well paced and rehearsed; each member presents a substantive technical portion and handles group questions confidently. The claimed division of labour matches what is observed in the viva.
Very good (70–84%)	Work shared fairly and documented, with some unevenness in timing or distribution. All members present competently and answer questions on their own contribution. Contribution statement broadly consistent with observed knowledge.
Pass (60–69%)	Contributions uneven or thinly evidenced; most work concentrated in the final weeks. Presentation delivered but under-rehearsed, or some members contribute little to it.
Fail (0–59%)	No credible evidence of planning or shared work; work log absent or fabricated; a member is absent from the demonstration, contributes nothing to the presentation, or the contribution statement is contradicted by the viva.

Individual marks

C1, C2, C4, C5, C6, C8, C9, C10 and C11 (70 marks) are awarded to the group. C3, C7 and C12 (30 marks) are awarded **individually**, so members of the same group will normally receive different totals.

A member who cannot explain work they claim to have produced may be awarded zero for the affected criteria.

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9. Report template

> Copy the structure below. Delete the guidance in italics. Target 2,000–3,000 words excluding references, tables, figures and appendices.

Title page — project title, course code and title, group number, member names and IDs, tutor name, submission date, word count.

Abstract *(150–200 words)* — problem, dataset, models, headline result, main conclusion.

Table of contents

1. Introduction

- 1.1 Problem statement and motivation *(what problem, why it matters, who benefits)*
- 1.2 Aim and objectives *(one aim; three to five measurable objectives)*
- 1.3 Scope and assumptions
- 1.4 Report structure

2. Background and related work

- 2.1 Application domain overview
- 2.2 Review of related approaches *(minimum three credible sources; what they did, what results, what limitation you address)*
- 2.3 Theoretical basis of the selected architecture(s) *(layer operations, activation functions, loss function, optimiser and training algorithm — this section and your viva carry CILO 1)*
- 2.4 Justification of the chosen approach

3. Dataset and preprocessing

- 3.1 Dataset description *(source, licence, size, classes, class distribution, sample examples)*
- 3.2 Exploratory data analysis *(at least two figures with interpretation)*
- 3.3 Preprocessing pipeline *(each step and why)*
- 3.4 Train / validation / test split *(sizes, stratification, leakage prevention)*

4. Methodology

- 4.1 Overall workflow *(a diagram is expected)*
- 4.2 Baseline model architecture *(layer table, parameter count)*
- 4.3 Improved / alternative model architecture
- 4.4 Training configuration *(loss, optimiser, learning rate, batch size, epochs, callbacks, hardware, seed)*
- 4.5 Hyperparameter tuning strategy *(what was varied and over what range)*

- 4.6 Evaluation metrics and justification

5. Results

- 5.1 Training behaviour *(loss and accuracy curves per model, with interpretation)*
- 5.2 Hyperparameter experiment table

Exp.	Parameter varied	Value	Val. accuracy	Val. loss	Observation
1					

- 5.3 Test-set performance

Model	Accuracy	Precision	Recall	F1	AUC	Params	Train time
Baseline							
Improved							

- 5.4 Confusion matrices and per-class results
- 5.5 Error analysis *(misclassified examples and probable causes)*

6. Discussion

- 6.1 Interpretation of results against the objectives
- 6.2 Model comparison and deployment recommendation
- 6.3 Interpretability findings *(Grad-CAM, saliency, attention, feature importance, embeddings — whichever fits your data)*
- 6.4 Limitations
- 6.5 Ethical, privacy and bias considerations

7. Conclusion and future work

References *(consistent style — [APA 7th / IEEE], as directed by your tutor)*

Appendices

- A. Contribution statement
- B. Work log / commit history
- C. Sources adapted, and what you changed *(list every tutorial, repository or documentation example your code draws on)*
- D. Additional figures and tables

Notebook structure template

```
1.0  Setup — imports, versions, random seed, device
2.0  Data loading
3.0  Exploratory data analysis
4.0  Preprocessing and splitting
5.0  Baseline model
     5.1 Architecture   5.2 Training   5.3 Curves
6.0  Improved model
     6.1 Architecture   6.2 Training   6.3 Curves
```

```
7.0  Hyperparameter experiments
8.0  Test-set evaluation — metrics, confusion matrix, per-class report
9.0  Interpretability and error analysis
10.0 Conclusion cell — summary of findings
```

Every section should open with a markdown cell explaining what the code below does and why. **Every code cell must be attributable to a named member** — add an author tag in the cell's opening comment. You will be asked about your cells by name.

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10. Contribution statement template

Group number: [ ] **Project title:** [ ]

Member name	ID	Main responsibilities	Report sections written	Notebook cells written	Presentation portion	Agreed share (%)	Signature

Shares must total 100%. All members sign to confirm the statement is accurate. Attach the work log or repository commit history as evidence. **This statement is checked against the individual viva** — claiming work you cannot explain is treated as academic misconduct.

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11. Project proposal template (Week [6], formative)

One page, submitted for tutor approval before work begins.

Field	Your entry
Group number and members	
Working project title	
Problem statement (2–3 sentences)	
Classification type	Binary / Multi-class / Multi-label
Dataset name, source and licence	
Approximate size and number of classes	

Field	Your entry
Candidate architectures (baseline + improved)	
Framework	TensorFlow / PyTorch
Planned evaluation metrics	
Anticipated challenges	
Role allocation	
Tutor approval	☐ Approved ☐ Revise and resubmit

12. Policies

Generative AI. Per the course descriptor, the use of generative AI tools in this assessment is **strictly prohibited unless explicitly permitted by the tutor**. Where the tutor grants permission for a specific purpose, you must declare the tool, the prompts used and the extent of the output retained, in an appendix. Undeclared use is treated as academic misconduct.

Note that 30 of the 100 marks are awarded live, individually, and cannot be prepared in advance. Submitting work you did not produce and cannot explain does not merely risk a misconduct finding — it also forfeits nearly a third of the available marks outright, which alone places the submission below the 60% pass mark.

Academic integrity. All code, text, figures and data adapted from other sources must be cited, including code from tutorials, repositories and documentation, and listed in Appendix C. Using a pre-trained model is permitted and encouraged where justified — presenting someone else's full pipeline as your own is not. Submissions may be checked for similarity.

Late submission. Late submissions will be treated as per the Academic Regulations of the Bahrain Polytechnic 2026-2027.

Reproducibility. If the notebook does not run, or results in the report cannot be located in the notebook outputs, the affected criteria will be marked on the evidence available only.

Pre-submission checklist

- ☐ Problem, dataset and objectives clearly stated and tutor-approved
- ☐ Three or more credible sources reviewed and cited
- ☐ Architecture theory explained, not just described
- ☐ Preprocessing pipeline complete and justified; no leakage between splits
- ☐ Baseline **and** improved model implemented and documented
- ☐ Three or more hyperparameters tuned, recorded in a table
- ☐ Learning curves included and interpreted
- ☐ Test-set metrics beyond accuracy, with confusion matrix and per-class results

- ☐ Models compared in one table with a reasoned recommendation
- ☐ Error analysis and interpretability section included
- ☐ Limitations and ethics discussed
- ☐ All figures numbered, captioned and referenced in the text
- ☐ Word count within limit; references consistent
- ☐ Notebook runs end to end with outputs saved; seed fixed; **every cell author-tagged**
- ☐ Appendix C lists every adapted source and what you changed
- ☐ README with library versions and reproduction steps
- ☐ Contribution statement signed by all members; work log attached
- ☐ ZIP named correctly and submitted before the Week 13 deadline

Before the demonstration

- ☐ Every member can explain **any** cell in the notebook, not only their own
- ☐ Environment installed and dataset available offline on the demo machine
- ☐ Each member has rehearsed making a small change to the notebook without help
- ☐ Presentation rehearsed with all members speaking to a technical portion